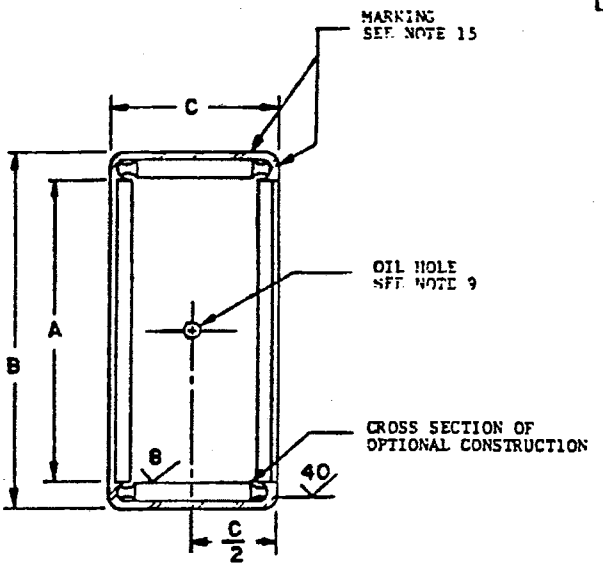


REVIEW ACTIVITIES: ARMY-ME, AV, MI; AF-99; PLA-15

USER ACTIVITIES: ARMY-AR, CL; NAVY-TD, MC, SH

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for reactivation use shall be made from this document when applicable.

APPROVED 30 APRIL 1980 REVISED (A) 25 SEP 1980

(NOTE 4) (NOTE 6) (NOTE 4) (NOTE 10)(NOTE 11) (NOTE 13)

MS PART NO.	A BORE			B DIA		C WIDTH +.000 -.010	RECOMMENDED SHAFT DIA		BASIC STATIC LOAD RATING LB	BASIC DYNAMIC LOAD RATING LB	APPROX LIMITING SPEED RPM
	NOM	MIN	MAX	RING GAGE	OUTSIDE NOM		MAX	MIN			
MS52142 -1	1/8	.1256	.1260	.2473	1/4	.250	.1251	.1248	221	236	12500
MS " -2	5/32	.1569	.1573	.2785	9/32	.312	.1564	.1561	357	374	10700
MS " -3	3/16	.1881	.1885	.3390	11/32	.250	.1876	.1873	263	309	10700
MS " -4	3/16	.1881	.1885	.3390	11/32	.375	.1876	.1873	575	537	10700
MS " -5	1/4	.2506	.2510	.4328	7/16	.312	.2501	.2498	462	495	10000
MS " -6	1/4	.2506	.2510	.4328	7/16	.438	.2501	.2498	865	786	10000
MS " -7	5/16	.3131	.3135	.4953	1/2	.312	.3126	.3123	578	573	8300
MS " -8	5/16	.3131	.3135	.4953	1/2	.438	.3126	.3123	1040	903	8300
MS " -9	3/8	.3756	.3760	.5578	9/16	.375	.3751	.3748	941	826	7100
MS " -10	3/8	.3756	.3760	.5578	9/16	.500	.3751	.3748	1430	1120	7100
MS " -11	7/16	.4381	.4385	.6203	5/8	.500	.4376	.4373	1690	1230	6300
MS " -12	1/2	.5006	.5010	.6828	11/16	.375	.5001	.4998	1210	958	5500
MS " -13	1/2	.5006	.5010	.6828	11/16	500	.5001	.4998	1900	1580	5500
MS " -14	1/2	.5006	.5010	.6828	11/16	.750	.5001	.4998	3280	2100	5500
MS " -15	9/16	.5631	.5635	.7453	3/4	.375	.5626	.5623	1340	1030	5000
MS " -16	9/16	.5631	.5635	.7453	3/4	.500	.5626	.5623	2110	1470	5000
MS " -17	9/16	.5631	.5635	.7453	3/4	.750	.5626	.5623	3640	2240	5000
MS " -18	5/8	.6256	.6260	.8078	13/16	.500	.6251	.6248	2320	1660	4500
MS " -19	5/8	.6256	.6260	.8078	13/16	.750	.6251	.6248	4010	2300	4500
MS " -20	11/16	.6881	.6885	.8703	7/8	.500	.6876	.6873	2530	1640	4200
MS " -21	11/16	.6881	.6885	.8703	7/8	.750	.6876	.6873	4370	4510	4200
MS " -22	3/4	.7503	.7507	.9950	1	.500	.7501	.7498	2630	2000	5300
MS " -23	3/4	.7503	.7507	.9950	1	.750	.7501	.7498	4700	3140	5300

(A) DENOTES CHANGES

P.A. AT Other Cust NAVY-OS AF-11	TITLE BEARING, ROLLER, NEEDLE: DRAWN OUTER RING, FULL COMPLEMENT, WITHOUT INNER RING, OPEN END, EXTRA PRECISION TYPE		MILITARY STANDARD MS52142
PROCUREMENT SPECIFICATION NONE	SUPERSEDES		SHEET 1 OF 3

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(NOTE 4) (NOTE 6) (NOTE 4) (A) (A) (NOTE 10) (NOTE 11) (NOTE 13)

MS PART NO.	A BORE NOM	A BORE MIN	A BORE MAX	F DIA RING GAGE	F DIA ROLLER SET NOM	C WIDTH +.000 -.010	RECOMMENDED SHAFT DIA MAX	RECOMMENDED SHAFT DIA MIN	BASIC STATIC LOAD RATING LB	BASIC DYNAMIC LOAD RATING LB	APPROX LIMITING SPEED RPM
MS52142 -24	13/16	.8126	.8132	1.0575	1-1/16	.500	.8126	.8123	2820	2090	5000
MS " -25	13/16	.8128	.8132	1.0575	1-1/16	1.000	.8126	.8123	7280	4360	5000
MS " -26	7/8	.8753	.8757	1.1200	1-1/8	.750	.8751	.8748	5400	3420	4500
MS " -27	7/8	.8753	.8757	1.1200	1-1/8	1.000	.8751	.8748	7780	4540	4500
MS " -28	15/16	.9378	.9382	1.1825	1-3/16	1.000	.9376	.9373	8280	4720	4400
MS " -29	1	1.0003	1.0007	1.2450	1-1/4	.750	1.0001	.9998	6100	3690	4100
MS " -30	1	1.0003	1.0007	1.2450	1-1/4	1.000	1.0001	.9998	8780	4900	4100
MS " -31	1-1/16	1.0628	1.0632	1.3075	1-5/16	.625	1.0626	1.0623	5020	3140	3800
MS " -32	1-1/8	1.1253	1.1257	1.3700	1-3/8	.750	1.1251	1.1248	6790	3930	3600
MS " -33	1-1/8	1.1253	1.1257	1.3700	1-3/8	1.000	1.1251	1.1248	9780	5220	3600
MS " -34	1-3/16	1.1878	1.1882	1.4950	1-1/2	.625	1.1876	1.1873	5100	3430	4400
MS " -35	1-1/4	1.2503	1.2507	1.4950	1-1/2	1.000	1.2501	1.2498	10800	5530	3300
MS " -36	1-1/4	1.2503	1.2507	1.4950	1-1/2	1.250	1.2501	1.2498	14100	6800	3300
MS " -37	1-5/16	1.3128	1.3132	1.6200	1-5/8	.500	1.3126	1.3123	3840	2710	4000
MS " -38	1-5/16	1.3128	1.3132	1.6200	1-5/8	.625	1.3126	1.3123	5590	3630	4000
MS " -39	1-3/8	1.3753	1.3757	1.6200	1-5/8	.750	1.3750	1.3747	8190	4370	3000
MS " -40	1-3/8	1.3753	1.3757	1.6200	1-5/8	1.250	1.3750	1.3747	15400	7140	3000
MS " -41	1-1/2	1.5003	1.5008	1.8700	1-7/8	.625	1.5000	1.4997	6200	4220	4300
MS " -42	1-1/2	1.5003	1.5008	1.8700	1-7/8	1.000	1.5000	1.4997	12300	7170	4300
MS " -43	1-1/2	1.5003	1.5008	1.8700	1-7/8	1.250	1.5000	1.4997	16300	8940	4300
MS " -44	1-5/8	1.6253	1.6258	1.9950	2	.625	1.6250	1.6247	6620	4350	3900
MS " -45	1-5/8	1.6253	1.6258	1.9950	2	1.250	1.6250	1.6247	17500	9260	3900
MS " -46	1-3/4	1.7503	1.7508	2.1200	2-1/8	.750	1.7500	1.7497	9400	5620	3600
MS " -47	1-3/4	1.7503	1.7508	2.1200	2-1/8	1.000	1.7500	1.7497	14000	7880	3600
MS " -48	1-3/4	1.7503	1.7508	2.1200	2-1/8	1.500	1.7500	1.7497	23500	11400	3600
MS " -49	1-7/8	1.8753	1.8758	2.2450	2-1/4	.500	1.8750	1.8747	5140	3500	3500
MS " -50	1-7/8	1.8753	1.8758	2.2450	2-1/4	1.000	1.8750	1.8747	15000	6080	3500
MS " -51	2	2.0003	2.0008	2.3700	2-3/8	1.000	2.0000	1.9997	15900	6300	3300
MS " -52	2	2.0003	2.0008	2.3700	2-3/8	1.750	2.0000	1.9997	31700	14200	3300
MS " -53	2-1/8	2.1253	2.1258	2.4950	2-1/2	1.000	2.1249	2.1246	16800	8510	3000
MS " -54	2-1/8	2.1253	2.1258	2.4950	2-1/2	1.500	2.1249	2.1246	28000	12600	3000
MS " -55	2-1/4	2.2503	2.2508	2.6200	2-5/8	.750	2.2499	2.2496	12000	6570	3000
MS " -56	2-1/4	2.2503	2.2508	2.6200	2-5/8	1.500	2.2499	2.2496	29600	13300	3000
MS " -57	2-1/2	2.5006	2.5012	2.8745	2-7/8	1.500	2.5000	2.4997	32700	13900	2700
MS " -58	2-5/8	2.6254	2.6260	2.9950	3	1.000	2.6248	2.6245	20600	9570	2500
MS " -59	2-3/4	2.7504	2.7510	3.1200	3-1/8	1.000	2.7498	2.7495	21600	9870	2500
MS " -60	2-3/4	2.7504	2.7510	3.1200	3-1/8	1.250	2.7498	2.7495	28700	12300	2500
MS " -61	2-1/2	3.5004	3.5010	3.9950	4	.750	3.4998	3.4995	16900	9270	2700

MATERIAL: ROLLERS: STEEL, ALLOY OR CARBON, E50100, E51100, E52100 OR 1090 OR 1095 PER FED-STD-66 OR SAE J-404.
OUTER RING: STEEL ALLOY OR CARBON, CARBURIZING GRADE, 4620, 4720, 8620, 8720 OR 1010-1020 PER FED-STD-66.

HEAT TREATMENT: NEEDLE ROLLERS: THROUGH HARDENED TO RC58 MIN.
OUTER RING BORE RACEWAY CASE HARDENED TO SURFACE HARDNESS RC58-65. CASE DEPTH .003 MIN.

SURFACE FINISH: SURFACES MARKED $\sqrt{8}$ AND $\sqrt{40}$ SHALL HAVE MAXIMUM SURFACE ROUGHNESS AS INDICATED, PER ANSI B46.1.

PROTECTIVE COATING: PLAIN (NOT PLATED). BEARINGS TO BE COATED WITH RUST PREVENTIVE FILM BY MANUFACTURER.

LUBRICATION: BEARINGS SHALL BE FURNISHED WITHOUT LUBRICATION.

APPROVED 30 APRIL 1980 REVISED (A) FOR CHANGES SEE SHEETS 2 AND 3

P.A. Other Cost	TITLE BEARING, ROLLER, NEEDLE: DRAWN OUTER RING, FULL COMPLEMENT, WITHOUT INNER RING, OPEN END, EXTRA PRECISION TYPE	MILITARY STANDARD MS52142
PROCUREMENT SPECIFICATION	SUPERSEDES:	SHEET 2 OF 3

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NOTES:

1. DIMENSIONS ARE IN INCHES.
2. ROLLERS SHALL BE RETAINED BY THE OUTER RING.
3. BEARINGS SHALL NOT BE FURNISHED WITH ROLLER SEPARATORS.
4. THE "GO" PLUG GAGE IS THE SAME SIZE AS THE MINIMUM DIAMETER UNDER THE NEEDLE ROLLERS, AND THE "NO GO" PLUG GAGE IS .0001 INCH LARGER THAN THE MAXIMUM DIAMETER UNDER THE NEEDLE ROLLERS. INSPECTION OF THE BEARING BORE SHOULD BE MADE WITH THE BEARING PRESSED INTO A RING GAGE OF THE SIZE SHOWN IN THE RING GAGE DIAMETER COLUMN.
5. BEARINGS ARE INTENDED TO BE INSTALLED ON SHAFTS WHERE MAXIMUM DEFLECTION DOES NOT EXCEED .0010 INCHES PER INCH OF BEARING WIDTH.
6. (A) BEARINGS ARE INTENDED TO BE USED WITH HARDENED SHAFTS RC58-65. WHEN USED WITH UNHARDENED SHAFTS, BEARING SHOULD BE USED IN CONJUNCTION WITH RING, BEARING, INNER, MS17130.
7. RECOMMENDED STEEL HOUSING BORE DIMENSIONS ARE AS FOLLOWS: RING GAGE DIAMETER PLUS .0000 MINUS 0.0003, FOR BEARING OUTSIDE DIAMETERS THROUGH 1; RING GAGE DIAMETER PLUS 0.0003 MINUS 0.0000, FOR BEARING OUTSIDE DIAMETERS 1-1/16 THROUGH 1-1/8; RING GAGE DIAMETER PLUS 0.0004 MINUS 0.0000, FOR BEARING OUTSIDE DIAMETERS 1-3/16 THROUGH 1-3/8; RING GAGE DIAMETER PLUS 0.0005 MINUS 0.0000, FOR BEARING OUTSIDE DIAMETERS 1-1/2 THROUGH 2-5/8; RING GAGE DIAMETER PLUS 0.0006 MINUS 0.0000, 2-7/8 AND GREATER.
8. MOUNTING IN CONFORMANCE WITH THE RECOMMENDED SHAFT DIAMETERS AND HOUSING BORE DIAMETERS RESULTS IN THE FOLLOWING DIAMETRAL CLEARANCES: .0002 TO .0012, FOR BEARINGS WITH BORE DIAMETERS UP TO AND INCLUDING 7/8; .0002 TO .0013, FROM 15/16 THROUGH 1-1/8; .0002 TO .0014, FROM 1-3/16 THROUGH 1-5/16; .0002 TO .0015, FOR 1-3/8; .0003 TO .0016, FROM 1-1/2 THROUGH 2; .0004 TO .0017, FROM 2-1/8 THROUGH 2-1/4; .0006 TO .0021, FROM 2-1/2 THROUGH 3-1/2.
9. OIL HOLE SHALL BE FURNISHED IN ACCORDANCE WITH MANUFACTURER'S STANDARD PRACTICE.
10. THE "BASIC STATIC LOAD RATING" FOR A RADIAL ROLLER BEARING IS THAT STATIC RADIAL LOAD WHICH CORRESPONDS TO A COMBINED TOTAL PERMANENT DEFORMATION OF ROLLER PLUS RACEWAY, AT THE MOST HEAVILY STRESSED CONTACT, OF .0001 OF THE ROLLER DIAMETER.
11. THE "BASIC DYNAMIC LOAD RATING" FOR A RADIAL ROLLER BEARING IS THAT CALCULATED, CONSTANT, RADIAL LOAD WHICH A GROUP OF APPARENTLY IDENTICAL BEARINGS WITH STATIONARY OUTER RING CAN THEORETICALLY ENDURE FOR A RATING LIFE OF ONE MILLION REVOLUTIONS OF THE SHAFT. THE RATING LIFE IS DEFINED AS THE NUMBER OF REVOLUTIONS THAT 90% OF A GROUP OF BEARINGS WILL COMPLETE OR EXCEED BEFORE THE FIRST EVIDENCE OF FATIGUE DEVELOPS. SINCE APPLIED LOADING AS GREAT AS THE BASIC DYNAMIC LOAD RATING TENDS TO CAUSE LOCAL PLASTIC DEFORMATION OF THE ROLLING SURFACES, IT IS NOT ANTICIPATED THAT SUCH HEAVY LOADING WOULD NORMALLY BE APPLIED.
12. FOR RATING LIVES AT LOADS OTHER THAN THE BASIC DYNAMIC LOAD RATINGS, AT OPERATING CONDITIONS OF THE INNER RING ROTATING, OUTER RING STATIONARY, STEADY LOAD, UNIFORM RPM, THOROUGH LUBRICATION, 300° F MAXIMUM BEARING TEMPERATURE, SHAFT MISALINEMENT NOT TO EXCEED .0010 INCHES PER INCH OF BEARING WIDTH.
13. LIMITING SPEED CALCULATED AT 300° F MAXIMUM, WITH THOROUGH LUBRICATION, AT BASIC LOAD RATING.
EMPIRICAL FORMULA FOR LOAD-LIFE RELATIONSHIP:

$$L = \left[\frac{Cr}{P}\right]^{\frac{10}{3}}, \text{ WHERE}$$

L = RATING LIFE, MILLIONS OF REVOLUTIONS.
Cr = BASIC DYNAMIC LOAD RATING, LB.
P = EQUIVALENT RADIAL LOAD TO WHICH BEARING IS SUBJECTED, LB.

FOR AN APPLIED LOAD GREATER THAN ONE-HALF THE BASIC DYNAMIC LOAD RATING, THE ABOVE LOAD LIFE RELATIONSHIP IS NOT VALID.

14. AVERAGE LIFE, THE LIFE WHICH 50% OF A GROUP OF BEARINGS WILL COMPLETE BEFORE EVIDENCE OF FATIGUE DEVELOPS, IS APPROXIMATELY FIVE (5) TIMES THE RATING LIFE.
15. MARKING SHALL CONSIST OF THE MS PART NUMBER AND THE MANUFACTURER'S IDENTIFICATION IN ACCORDANCE WITH MIL-STD-130.
16. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATION FOR BIDS OR REQUEST FOR PROPOSAL, EXCEPT THAT REFERENCED ADOPTED INDUSTRY DOCUMENTS SHALL GIVE THE DATE OF THE ISSUE ADOPTED.
17. FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN.
18. THE USE OF RECYCLED MATERIALS WHICH MEET THE REQUIREMENTS OF THE APPLICABLE MATERIAL SPECIFICATION WITHOUT JEOPARDIZING THE INTENDED USE OF THE ITEM SHALL BE ENCOURAGED.

APPROVED 30 APRIL 1980 REVISED (A) FOR CHANGES SEE SHEETS 2 AND 3

P.A. Other Cost	TITLE BEARING, ROLLER, NEEDLE: DRAWN OUTER RING, FULL COMPLEMENT, WITHOUT INNER RING, OPEN END, EXTRA PRECISION TYPE	MILITARY STANDARD MS52142
PROCUREMENT SPECIFICATION	SUPERSEDES:	SHEET 3 OF 3

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